DEVELOPER ROLE

SET C — 100 MCQs

Practice assessment · Java, DSA, SQL/DBMS, web, Spring Boot, security, Git/testing, OS and system design.

Instructions: Select one best answer per question. Suggested time: 75 minutes. One mark per question; no negative marking in this practice version.

1. Which SQL operator checks whether a value matches any value in a list?

A. BETWEEN
B. LIKE ONLY
C. IN
D. IS NULL

2. What is method overriding?

A. Subclass implementation of an inherited method
B. Calling a static method
C. Changing a variable type
D. Declaring multiple constructors

3. What does JVM stand for?

A. Java Virtual Machine
B. Java Verified Module
C. Java Variable Manager
D. Joint Version Model

4. Which key uniquely identifies a table row?

A. Foreign key
B. Primary key
C. Secondary index
D. Duplicate key

5. What is quicksort's worst-case time complexity?

A. O(n)
B. O(n log n)
C. O(log n)
D. $O(n^2)$

6. Which is a checked Java exception?

A. ArrayIndexOutOfBoundsException
B. ArithmeticException
C. FileNotFoundException
D. NullPointerException

7. Which HTTP status commonly means resource created?

A. 301
B. 200
C. 404
D. 201

8. Which interface is the root of the Java collection hierarchy for groups of elements?

A. Comparable
B. Collection
C. Runnable
D. Serializable

9. Which SQL statement changes existing rows?

A. GRANT
B. ALTER
C. UPDATE
D. CREATE

10. What does HTTPS add to HTTP?

A. Mandatory compression
B. TLS transport security
C. Automatic database caching
D. Guaranteed zero latency

11. Which application layer usually contains business rules?

A. DNS
B. Presentation
C. Physical network
D. Service/business

12. Binary search on a sorted array takes what worst-case time?

A. $O(n^2)$
B. O(1)
C. O(log n)
D. O(n)

13. Which data structure follows FIFO?

A. Set
B. Queue
C. Tree
D. Stack

14. What is SQL injection?

A. Attack manipulating SQL via untrusted input
B. Backup method
C. Normalization form
D. Query optimization

15. What does static mean for a class field?

A. One copy per object
B. The field is always immutable
C. A class-level member shared by instances
D. The field cannot be read

16. Which OOP principle hides internal state behind methods?

A. Polymorphism
B. Abstraction
C. Encapsulation
D. Inheritance

17. What is a database transaction?

A. File format
B. Index type
C. Logical unit of work committed or rolled back
D. User account

18. What is the purpose of memoization?

A. Increase recursion depth
B. Sort linked lists
C. Encrypt variables
D. Cache results of repeated function calls

19. What does least privilege mean?

A. Disable authentication
B. Grant only necessary permissions
C. Share one account
D. Everyone gets admin access

20. What does a generic type parameter provide?

A. Automatic persistence
B. Thread scheduling
C. Runtime encryption
D. Compile-time type safety

21. Which technique explores a graph level by level?

A. BFS
B. Heapify
C. DFS
D. Backtracking only

22. Which Java collection rejects duplicate elements?

A. Vector
B. HashSet
C. ArrayList
D. LinkedList

23. Normalization primarily reduces what?

A. Redundancy and update anomalies
B. All joins
C. Query syntax
D. Indexes

24. What is the purpose of try-with-resources?

A. Create threads
B. Suppress all exceptions
C. Automatically close AutoCloseable resources
D. Retry failed requests

25. Which status code represents an internal server error?

A. 302
B. 404
C. 500
D. 201

26. What is Git primarily used for?

A. Web design
B. Running production servers
C. Relational storage
D. Version control and collaboration

27. Which Java type stores key-value pairs?

A. HashSet
B. PriorityQueue
C. HashMap
D. ArrayList

28. What does a repository usually abstract?

A. Browser history
B. DNS configuration
C. Data access operations
D. CSS rendering

29. Which command removes all rows while retaining table structure?

A. DROP
B. TRUNCATE
C. RENAME
D. ALTER

30. Which algorithm finds shortest paths in an unweighted graph?

A. BFS
B. Binary search
C. Bubble sort
D. DFS

31. What is the default value of an instance int field in Java?

A. 0
B. null
C. Uninitialized random value
D. -1

32. What is true about Java interfaces?

A. A class can extend multiple classes
B. Interfaces require constructors
C. A class can implement multiple interfaces
D. Interfaces cannot have default methods

33. Which standard Java class is immutable?

A. StringBuilder
B. String
C. StringBuffer
D. ArrayList

34. Dynamic programming is especially useful when subproblems are what?

A. Overlapping with reusable results
B. Always independent
C. Limited to graphs
D. Only sorted

35. What does COUNT(*) count?

A. Only non-null values in one column
B. Only primary keys
C. Only distinct rows
D. Rows in the result set

36. Which layer should normally handle HTTP request mapping?

A. Controller
B. Database index
C. Build tool
D. Repository

37. What does GROUP BY do?

A. Renames a table
B. Groups rows for aggregation
C. Sort alphabetically
D. Deletes duplicates

38. Which map implementation generally does not guarantee iteration order?

A. LinkedHashMap
B. TreeMap
C. EnumMap
D. HashMap

39. Which keyword explicitly throws an exception?

A. catch
B. throw
C. throws
D. final

40. What does DNS do?

A. Encrypt HTTP traffic
B. Resolve domain names to IP addresses
C. Compile Java
D. Create database keys

41. Which SQL command retrieves rows?

A. GET
B. OPEN
C. SELECT
D. FETCH TABLE

42. Which Java type is designed for mutable strings without synchronization?

A. StringBuilder
B. StringLiteral
C. String
D. Character

43. What is a constructor used for?

A. Destroying an object
B. Handling exceptions only
C. Starting a thread automatically
D. Initializing a new object

44. Which constraint prevents NULL values in a column?

A. CHECK
B. FOREIGN KEY
C. DEFAULT
D. NOT NULL

45. Which HTTP status means authentication is missing or invalid?

A. 400
B. 500
C. 401
D. 403

46. Which index structure is common for database range queries?

A. Call stack
B. B-tree/B+ tree
C. Stack
D. Binary heap

47. Insert at the head of a linked list when head is known takes?

A. O(1)
B. O(n)
C. O(log n)
D. $O(n^2)$

48. What does git pull typically do?

A. Rewrites all commit messages
B. Fetches and integrates remote changes
C. Creates a hosted repository
D. Deletes local branches

49. What is a DTO used for?

A. Transfer data across application boundaries
B. Replace every service
C. Start threads
D. Exact database engine representation

50. A balanced BST typically supports search in what time?

A. O(1) guaranteed
B. O(log n)
C. O(n²)
D. O(2■)

51. What is the purpose of a database index?

A. Encrypt every row
B. Replace primary keys
C. Speed up selected reads at storage/write trade-off
D. Prevent all duplicates automatically

52. What does a JWT signature provide?

A. Infinite token lifetime
B. Integrity and authenticity verification
C. Confidentiality of all claims
D. Password storage

53. Which file commonly holds Spring Boot configuration?

A. index.css
B. routes.sql
C. application.properties
D. pom.lock

54. Which technique is commonly used to find connected components?

A. Insertion sort
B. Binary search only
C. Hashing strings only
D. Graph traversal

55. Why paginate API results?

A. Encrypt payloads
B. Prevent all errors
C. Return large result sets in manageable chunks
D. Replace authorization

56. Which command changes a table definition?

A. SELECT
B. ALTER TABLE
C. COMMIT
D. UPDATE

57. Which join returns only matching rows from both tables?

A. FULL OUTER JOIN
B. LEFT JOIN
C. INNER JOIN
D. CROSS JOIN

58. Which access modifier gives the broadest visibility?

A. public
B. package-private
C. protected
D. private

59. Which block commonly runs whether or not an exception occurs?

A. finally
B. catch only
C. throw
D. assert

60. What does a foreign key enforce?

A. Encryption
B. Referential integrity
C. Compression
D. Sorting

61. Which HTTP method is commonly used for partial updates?

A. GET
B. TRACE
C. PATCH
D. OPTIONS

62. Which Java keyword prevents class inheritance?

A. final
B. private
C. volatile
D. static

63. Which HTTP method is generally used to retrieve a resource?

A. DELETE
B. POST
C. PATCH
D. GET

64. What does CORS govern?

A. CPU scheduling
B. Database migrations
C. Cross-origin browser requests
D. Password hashing

65. Which algorithm finds a minimum spanning tree?

A. KMP
B. Binary search
C. Kruskal's algorithm
D. Floyd cycle detection

66. What is the typical average time complexity of hash-table insertion?

A. O(n log n)
B. O(1)
C. $O(n^2)$
D. O(log n)

67. Which Spring annotation marks a REST controller?

A. @ServletTable
B. @RestController
C. @AutowiredOnly
D. @EntityManager

68. What is Flyway commonly used for?

A. Bytecode compression
B. UI icons
C. Versioning and applying database schema changes
D. Load balancing

69. Average lookup in a well-distributed hash table is typically?

A. O(1)
B. O(n log n)
C. O(n)
D. O(log n)

70. An adjacency-list graph representation typically uses what space?

A. $O(V^2)$ always
B. $O(E^2)$
C. O(V + E)
D. O(log V)

71. What is method overloading?

A. Hiding a field
B. Subclass replacement of a method
C. Same name with different parameter lists
D. Changing a return value only

72. Which Spring annotation maps GET requests?

A. @GetMapping
B. @PutMapping
C. @PostMapping
D. @DeleteMapping

73. What is dependency injection?

A. Encrypting method names
B. Supplying an object's dependencies externally
C. Creating every dependency inside a class
D. Converting Java to SQL

74. Which data structure is suitable for union/find connectivity queries?

A. Stack
B. Deque only
C. Trie
D. Disjoint-set union

75. Which ACID property means all-or-nothing?

A. Consistency
B. Isolation
C. Atomicity
D. Durability

76. When properly overridden, equals() usually checks what?

A. Class loading order
B. Object size
C. Logical equality
D. Reference identity only

77. How should passwords be stored?

A. Reversible Base64
B. Salted slow hash such as BCrypt/Argon2
C. Unsalted MD5
D. Plain text

78. What does @Transactional help manage?

A. Database transaction boundaries
B. JWT keys only
C. HTML layout
D. Network routes

79. REST is best described as what?

A. CSS framework
B. Architectural style for networked applications
C. Database engine
D. Programming language

80. Which collection preserves insertion order and allows duplicates?

A. ArrayList
B. HashSet
C. TreeSet
D. HashMap

81. What helps prevent SQL injection?

A. Parameterized queries
B. Exposing SQL errors
C. Disabling indexes
D. Concatenating user input into SQL

82. Where should production secrets be kept?

A. Public README
B. Secret manager/protected environment
C. Committed source code
D. Frontend bundle

83. Which command creates and switches to a branch?

A. git stash pop
B. git fetch --delete
C. git reset --hard
D. git switch -c branch-name

84. Which protocol provides reliable ordered byte-stream delivery?

A. TCP
B. ICMP
C. ARP
D. UDP

85. Which structure efficiently retrieves the highest-priority item?

A. Singly linked list
B. Priority queue
C. Basic queue
D. Stack

86. What is the time complexity of merging two sorted arrays of total length n?

A. O(log n)
B. O(1)
C. $O(n^2)$
D. O(n)

87. How can a linked-list cycle be detected with O(1) extra space?

A. Counting sort
B. Binary search
C. Topological sort
D. Floyd slow/fast pointers

88. Which data structure follows LIFO?

A. Stack
B. Heap
C. Graph
D. Queue

89. Which clause filters rows before grouping?

A. HAVING
B. WHERE
C. ORDER BY
D. GROUP BY

90. Which keyword refers to the current Java object?

A. this
B. current
C. self
D. super

91. Which clause filters aggregated groups?

A. WHERE
B. HAVING
C. DISTINCT
D. LIMIT

92. What is the Java application entry-point method?

A. init()
B. run()
C. public static void main(String[] args)
D. start()

93. Which algorithmic strategy chooses a locally optimal option at each step?

A. Dynamic linking
B. Hashing
C. Normalization
D. Greedy

94. Which media type represents JSON?

A. image/java
B. application/java
C. application/json
D. text/sql

95. Which BST traversal yields sorted keys?

A. Postorder
B. Reverse level order
C. Preorder
D. Inorder

96. Which structure is commonly used for efficient prefix searches?

A. Stack
B. Queue
C. Trie
D. Binary heap

97. What does COMMIT do?

A. Persist transaction changes
B. Create an index
C. Delete a database
D. Undo a transaction

98. Which keyword makes a variable's visibility changes observable across threads but does not make compound operations atomic?

A. transient
B. volatile
C. strictfp
D. native

99. Which structure supports the call stack used by recursion?

A. Queue
B. Stack
C. Hash table
D. Disjoint set

100. Which sorting algorithm is typically stable?

A. Heap sort
B. Selection sort
C. In-place quicksort
D. Merge sort